

Chapter 1

Introduction

1.1 Need

Current steganography project is having complexity problem in terms of time per Data. Now Days Steganography project can hide data in equal length of image file[1-9], For Example It can hide 1Mb data with At least (Or More than) 1Mb Image file, it means that For Extracting 1Mb data we have to download At least (OR more than) 2Mb file (i.e. 50% loss of Data OR more than 50% loss), and also, its cost of transmission is more. Second problem is, it takes more time for a complete transmission from sender to receiver.

1.2 Basic Concept

LZW compression scheme is used to optimize the size of secret data, it will enable a person to hide approx two times more data in a cover-image[7][10], i.e. Now we can store Double Amount of Data into Same Image File. This Way we can reduce the cost of data and we can also reduce loss of data up to 50%. This approach is secure against the RS detection attack and its steno-image is totally indistinguishable from the original image (cover-image) by the human eye, and by using Hybrid Approach we can reduce transmission time by Extracting data parallely at decoding time. This way we can reduce some amount of transmission time.

1.3 Applications

- Confidential communication and secret data storing
- Access control system for digital content distribution
- We can use in media Database systems so that database will be secure.
- Can be use in every online data transfer.

Chapter 2

Literature Survey

Paper 1:

Intisar Majeed Saleh, Hanna Hameed Merah “Efficient Data Hiding System using LZW, Cryptography and GIF Image Steganography”, International Journal of Technical Research and Applications, 2015.

The combination of steganography and cryptography is considered as one of the best security methods used for message protection, due to this reason, in this paper, a data hiding system that is based on image steganography and cryptography is proposed to secure data transfer between the source and destination. Animated GIF image is chosen as a carrier file format for the steganography due to a wide use in web pages and a LSB (Least Significant Bits) algorithm is employed to hide the message inside the colors of the pixels of an animated GIF image frames. To increase the security of hiding, each frame of GIF image is converted to 256 color BMP image and the palette of them is sorted and reassign each pixel to its new index, furthermore, the message is encrypted by LZW (Lempel _ Ziv_Welch) compression algorithm before being hidden in the image frames. The proposed system was evaluated for effectiveness and the result shows that, the encryption and decryption methods used for developing the system make the security of the proposed system more efficient in securing data from unauthorized users. The system is therefore, recommended to be used by the Internet users for establishing a more secure communication.

Paper 2:

Aiswarya Baby, Hema Krishnan “Combined Strength of Steganography and Cryptography”, International Journal of Advanced Research in Computer Science, 2017.

The use of internet for communication purpose has rapidly increased and it magnified the attacks to users. Protecting the data is a big challenge for computer users. Cryptography

and Steganography are widely used techniques to ensure security. Both techniques have many applications in computer science and other related fields. Both methods provide security in their own ways, but to add multiple layers of security it is always a good practice to use combination of these techniques. The concepts of steganography, cryptography and their applications in the security of digital data communication across network is studied in this paper and technical survey of recent methods which combined steganography and cryptography is presented.

Paper 3:

Assit.Lec. May H.Abood “An Efficient Cryptography Using HASH-LSB Steganography with RC4 and Pixel Shuffling Encryption Algorithms”, Annual Conference on New Trends in Information & Communications Technology Applications-(NTICT'2017).

This paper improves information security through developing efficient image cryptography algorithm by using encryption with steganography. The proposed algorithm ensures the encryption and decryption using RC4 stream cipher and RGB pixel shuffling with steganography by using hash-least significant Bit (HLSB) that make use of hash function to developed significant way to insert data bits in LSB bits of RGB pixels of cover image. The security evaluations are presented by calculating a peak signal to noise ratio and mean square error. For secret image, PSNR is infinity and MSE is 0. For cover image, PSNR is about 63 dB and MSE is about 0.03. The results show that high level of the similarity exists between the stage-images and cover images and the same is for secret images and extracted image as represented also in In Histogram Analysis of secret images. These algorithms are performed by using MATLAB program.

Chapter 3

Existing System

They proposed new approach wherein both cryptography and steganography are used[1-7] to encrypt the data and also to conceal these encrypted contents in some other medium. In this method, we can secure Message File by converting it into an encrypted text using S-DES algorithm (Or Triple DES, AES, RSA, etc.)[9][6] and a secret key and then concealing this encrypted text in some other image. This technique has been tested and it has been observed that they prevent the possibilities of steganalysis also. Now days project of combination of steganography and cryptography either with LZW compression method or with single hybrid approach techniques. In case one they used combine strength of steganography and cryptography with alphabets repetition to compress the file because of this we can save the data to transfer a file over the internet . Average compression of message file is 50% (i.e. just half) And in case two, this also contain the combination of steganography and cryptography but only difference is instead of LZW compression algorithm, it use hybrid approach techniques[6][9]. Its mean during the extraction of message at receiver end it divides virtually to the whole stego-image file as in two part to decode a message file simultaneously from stego-image file. It saves time of decoding up to 50% of total time of decoding.

Chapter 4

Aim and Objective

4.1 Aim

The combine the strength of steganography and cryptography for a secure data and data hiding system that is based on image steganography and cryptography is proposed to secure data transfer between the source and destination. And also the main focus of this project is to combine the all different trends which are already been proposed in the direction of cryptography and steganography to reduce the time and space during transmission over the network.

4.2 Objective

- To study the security mechanism in currently existing system.
- To study and develop the module required for steganography to make the appearance of secret data invisible to any outside intruder.
- To study and develop the algorithms that can be used for easy encryption and decryption with image steganography so that we can reduce complexity..
- To study and develop the modules that can be used to reduce the time and space complexity during transmission over the network.
- To study and develop the modules that can used for lossless compression and decompression method.
- Reduce the transmission time by reducing decoding time as well as including data compression method.

Chapter 5

Problem Statement

Current steganography project is having complexity problem in terms of time per Data[1-9]. Now Days Steganography project can hide data in equal length of image file, For Example It can hide 1Mb data with At least (Or More than) 1Mb Image file, it means that For Extracting 1Mb data we have to download At least (OR more than) 2Mb file (i.e. 50% loss of Data OR more than 50% loss)[7][10]. And also, its cost of transmission is more. Second problem is, it takes more time for a complete transmission from sender to receiver. Many different methods are also proposed to nullify the above problem. But the solution of that is not completed. It means they haven't succeeded 100%. Use of LZW algorithm to reduce the file size, currently we are using compression in alphabets only, since LZW compression method works on repetition of words, its mean more the repetition less the size of compressed data file. But still as compare to repetition of words in ASCII value in more than repetition in alphabets. And also extraction of data from stego-image takes time to extract data. By using multi-threading approach, we can reduce decoding time.

Chapter 6

Proposed System for Project

The combination of steganography and cryptography is considered as one of the best security methods used for message protection, due to this reason, in this paper, a data hiding system that is based on image steganography and cryptography is proposed to secure data transfer between the source and destination. Animated GIF image is chosen as a carrier file format for the steganography due to a wide use in web pages and a LSB(Least Significant Bits) algorithm is employed to hide the message inside the colors of the pixels of an animated GIF image frames. To increase the security of hiding, each frame of GIF image is converted to 256 color BMP image and the palette of them is sorted and reassign each pixel to its new index, furthermore, the message is encrypted by LZW (Lempel _ Ziv_Welch) compression algorithm before being hidden in the image frames. The proposed system was evaluated for effectiveness and the result shows that, the encryption and decryption methods used for developing the system make the security of the proposed system more efficient in securing data from unauthorized users. The system is therefore, recommended to be used by the Internet users for establishing a more secure communication.

The main objective of the proposed system is to secure data from the unauthorized user by putting multiple layers of security on data. All layers should be in proper order so that system will work smoothly. The proposed method is using LSB technique to hide the data or embedding data into an image file. And also cryptography use to make more complex to understand actual information for unauthorized users. To increase hiding capacity of data into image file we use Lempel-Ziv-Welch (LZW) compression technique. It increases the hiding capacity of an image by 50%[7][10]. Now we can hide approx double data compare to the previous case. We are using a key which helps to improve prevention of unauthorized access of the system. Key helps sender to lock the stego-image during transmission. In this proposed system stego-image always contain a small code which will ask for a key to the user at extraction time, without right key user cannot extract information

from stego-image. At the extraction time, the proposed method is using multi-threading technique to retrieving data rapidly. Depending on number multiple threads extraction time will decrease by half each time[5]. This way we can extract data comparatively in less time so that we can increase the efficiency of the proposed system.

Chapter 7

Requirement Analysis

7.1 Introduction:

7.1.1 Purpose

The proposed system is a product which is capable of hiding the information into other information. The hidden information can be a text file, image file or simply a message. The Carrier file can one of image file. The product (software) should also support cryptography and compression with hybrid approach techniques to improve security level, reduce size of file and also reduce time of decoding(extraction of message file from stego-file) at the receiver end.

7.1.2 Document Conventions

Carrier File/Object: A Carrier file or Carrier object is the source file onto which the data to be hide is written in such a way the output file resembles the initial source file to naked eye and to the normal applications which are associated to the source file. For example, a plain file is generally edited in notepad. If we write some message in some way into the source file and produced an output file, the content of the file should be same as before when the output file is opened in notepad.

Secret File/Object: This object can be message, image file, text file. It is intended to be hidden in the Carrier File.

Hidden File/Object: it is same as Secret File/Object.

Output File: This is the file produced as a result of steganographic operation. It should resemble the Source file.

7.1.3 Intended Audience and Reading Suggestions

End Users: end users are the ones who will be using software the most. This SRS clearly suggests the requirements of the software. So, It is useful to the End Users as well to know what are the requirements and recommendations of the software, although all the requirements are restated into the User Manuals and Installation GUIDE as well.

Developers: The SRS is also useful to the developers who wish to develop their own implantations of Steganography. They can use SRS as directly to enhance the implemented Software or may develop a new Software right from scratch.

Testers: Because the software is bulky and large in size, all the testers testing the Steno-magic software should read SRS carefully to go through all the paths of the software during testing.

Students: The SRS is also knowledgeable for the students to get the information about Steganography. They can refer our SRS, SDD and Project Report to get an overview of Steganography.

7.2. Overall Description:

7.2.1 Product Perspective

Steganography is a very old technique of hiding the data. This software is all about hiding the data. This software is made according to the modern need of hiding data. It uses various new techniques for hiding the data. The basic advantage of this product is that it is not specific for a particular type of either hidden file or carrier file.

7.2.2 Product Features

- Carrier file can be text file, audio file, video file, and image file
- Various file format is supported
 - Text – txt, rtf, ham, html
 - Image- bmp, gif
- Hidden object can be a text file, video file, audio file, image file
- Hidden files can be of any format
- Encryption is supported
- Compression is supported
- Multiple hidden files can be stored in a single carrier file if the size permit
- Authentication is provided
- Authentication management is also there
- Past steganographic task can be viewed by the help of log file
- Scheduler is also available
- Look of the software can be changed at runtime
- Context sensitive is present
- User manual, installation guide, help file are also provided with the product

7.2.3 User Classes and Characteristics

Administrator: They have full control over the software. Apart from using the basic task, they have full control over the user management and they can also view the log file.

End User: They can perform the Steganographic task but have no control over the user management and log file.

7.2.4 Operating Environment

Operating System: Win XP, 2000, 98, Vista

Software req.: JRE 1.6 or higher,

Windows media player Recommended Conf.: 256mb RAM or higher,
10mb Disk space
Screen Resolution: 1024x768

7.2.5 Design and Implementation Constraints

Although java is portable, the software does not run properly on platforms other than windows. It cannot be run over LAN or internet. The software does not check the size limit. As a result, there may be a case that hidden object is not fully consumed yet the software shows no error. This can be verified by extracting the hidden object just after hiding it into the carrier file. Although compression is supported but not implicitly, so manual intervention is needed. Output audio file is distorted considerably when both carrier and hidden objects are audio.

7.2.6 User Documentation

Installation Guide, User Manual and Help File are provided separately with the product. Context sensitive help is integrated implicitly with the product.

7.2.7 Assumptions and Dependencies

Dependencies:

- Mp3plugin.jar is required to play the mp3 files implicitly by the software.
- Windows media player is required to present in the system.
- The Project Location is C:\Package\Stego\. If project is moved to different location the project may not run correctly.

Assumptions:

- The Operating System is any one of NT Family
- Media player is present in the system.
- Class path is set properly

- Mp3plugin.jar is present in the folder C:\Package\Stego

7.3. External Interface Requirements: -

7.3.1 User Interfaces

Splash Screen: It is the first screen which is shown to user. It loads other modules of the project.

Login Screen: It is for security purpose. It asks the user to give its name and password. Only authenticated user can use the software.

Main Form: It provides various features to user about the mode of use of software. User can

choose text, audio, video, image option and can do work on it. It also provides various other features like user management, compression, encryption, zip, log file.

Help: By clicking on help button help on corresponding topic is shown.

Back: By clicking on this button the main form will appear.

Exit: Click on exit button on main form will shut down the application.

Error Messages: Proper error messages will appear when any error is encountered.

7.3.2 Hardware Interfaces

Apart from the recommended configuration no other specific hardware is required to run the software.

7.3.3 Software Interfaces

The JRE is required to run the software. The JRE version should be 1.6 or higher. The only other software required is the Windows media player to play the video files. The mp3 plugin is also required to play mp3 files.

7.4 Functional Requirements:

- Image file is taken as in .jpeg or .gif format.

- The encrypted text file is compressed using LZW compression method.
- Compressed text file is converted into bit stream.
- The bit stream is then embedded into image file using LSB algorithm.
- At the receiver end de-steganography is carried using reverse LSB algorithm.
- The received bit stream is converted into text file then its decompress into encrypted text file.
- Then encrypted text file decrypted.

7.5. Nonfunctional Requirements:

7.5.1 Performance Requirements

The ram should be 128mb at least. But 256mb ram is recommended. The disk space required to store the software is 10mb and to store the output files and other configuration files associated with the software the recommended disk space required is 15mb.

7.5.2 Safety Requirements

The size constraints have to be evaluated by the end user only. The software does not check for the size constraint. This is done to enhance the performance of the software in terms of speed. In case size of hidden object exceeds the max. Allowable size that can be hidden, the extra information is truncated. The part of hidden object of size equal to maximum possible size is stored in carrier file. The size of hidden object that can be stored in a carrier file depends on the carrier file size and type of steganographic task. The screen resolution should be set to 1027x768 or higher to get the complete view of the software. In case of lower screen resolution, the software not only looks awkward but also not completely visible.

7.5.3 Security Requirements

The user must have a registered account to run the software. The administrator account is needed tube created on the first run of the software. Then the administrator account can be

used to create other user accounts. If a user forgets the password, the password can be retrieved on the basis of the Date of Birth. This date of birth is stored at the time of creation of the account although it can be modified later.

7.6. Other Requirements

The class path should be set properly. Otherwise the software will automatically set the class path but that will take time, due to which the performance decreases considerably. There are no further requirements other than the specified in this SRS under different headings.

Chapter 8

Scope

In our randomized LSB technique we have embedded two bits in a pixel using a message dependent randomized approach. In future we would like to exploit the possibility of hiding three bits in moderate bit locations in a randomized manner without disturbing the least significant bit in each pixel. The possibility of embedding by changing every pixel value with a new one which will conceal the hidden data can also be exploited. LSB substitution methods give high capacity. By using better encryption algorithm we can improve the security level and also using better lossless compression algorithm ,we can compress file as much we can.

8.1. Technical Feasibility

It determines the technology needed for the proposed system is available and how this technology can be integrated into the organization. Technical evolution must also assess whether the existing system can be upgraded to use the new technology and whether the organization has the expertise to use it.

8.2. Economical Feasibility

The economic feasibility of the system looks upon the financial aspects of the system. It determines whether is economically feasible or not. In other words, it determines whether the investment that goes into the implementation of the project is recoverable or not. The cost benefit analysis is the commonly used method in evaluating the effectiveness of the system. As the hardware is already available and no investment is to be made in that direction, the only cost involved is that of implementing the system and software.

8.3. Operational Feasibility

It considers whether users will be adapted to the system easily after training, can the system provide them a user friendly environment etc. Acceptance of the system by the user avoiding user resistance is considered as the main factor.

8.4. Cultural Feasibility:

This refers to the possible effects on end users after the release of the system or the acceptance of the system by the users observed while different stages of securities. The proposed system is an improvement over the existing system and also the system is equipped with an interactive graphical user interface which will ensure ease of use and hence the users who are currently operating with the existing systems will be able to adapt very easily with the new system. Also, the availability of online and offline help will lead to wide user acceptance as new novice users can easily be trained using this material.

8.5. Schedule feasibility:

This type of feasibility is there to ensure that the project is completed in time. The proposed system being an add-on to the existing system is implemented in a modular way hence exhibiting low coupling and a good modular independence because of which each module can be developed parallelly to a certain extent along with regular testing of the module. Thus, schedule slippage can easily be avoided because of the modular nature of the project

Chapter 9

Methodology

In this Proposed System, The general scheme for embedding data is depicted in Fig 9.1. A message is embedded in a file by the stego-system encoder, which has as inputs the original cover, the secret message and a key. The resulting stego-object is then transmitted over a communication channel to the recipient where the stego-system decoder using the same key processes it and the message can be read[1-7].

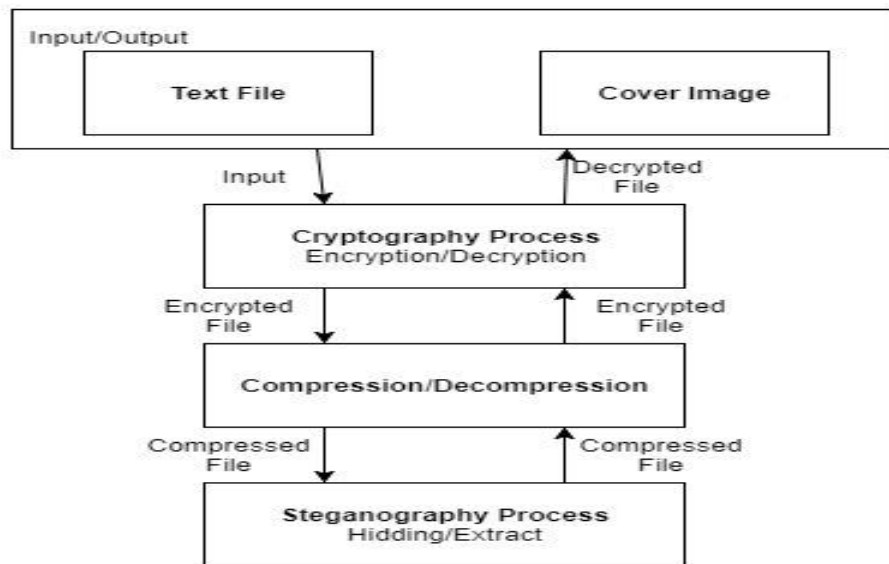


Fig 9.1. A General Steganographic Model

The steganographic process can be represented using formulas. The stego-object is given by:

$$I' = f(I, m, k)$$

where: I' is the stego-object, I is the original object, m is the message and k is the key that the two parties share. The stego-object may be subject to many distortions, which can be represented as a noise process n :

$$I'' = I' + n(I')$$

At the decoder we wish to extract the signal m , so we can consider the unwanted signal to be l . The embedded signal should resist common signal distortions as those depicted in Fig 9.2. Two kinds of compression exist: lossy and lossless. Both methods save storage space but have different results. Lossless compression permits exact reconstruction of the original message; therefore it is preferred when the original information must remain intact. Such compression schemes are the images saved as GIF (Graphic Interchange Format). Lossy compression, on the other hand, does not maintain the original's integrity. Such compression scheme is an image saved as JPEG (Joint Photographic Experts Group). The JPEG formats provide close approximations to high-quality digital photos but not an exact duplicate.

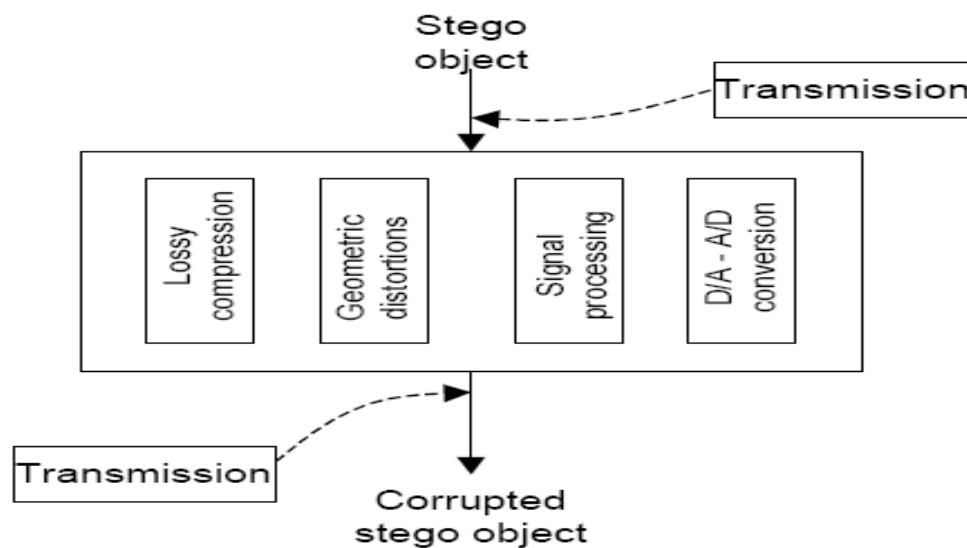


Fig 9.2: Common signal distortions over the transmission channel

9.1 Lest Significant Bit (LSB):-

Least significant bit (LSB) insertion is a common, simple approach to embedding information in a cover image [5]. In this method, we can take the binary representation of the hidden_data and overwrite the LSB of each byte within the cover_image. If we are using 24-bit color, the amount of change will be minimal and indiscernible to the human eye. As an example, suppose that we have three adjacent pixels (nine bytes) with the following RGB encoding:

```

10010101 00001101 11001001
10010110 00001111 11001010
10011111 00010000 11001011

```

Now suppose we want to "hide" the following 9 bits of data (the hidden data is usually compressed prior to being hidden): 101101101. If we overlay these 9 bits over the LSB of the 9 bytes above, we get the following (where bits in bold have been changed):

```

10010101 00001100 11001001
10010111 00001110 11001011
10011111 00010000 11001011

```

Note that we have successfully hidden 9 bits but at a cost of only changing 4, or roughly 50%, of the LSBs. Similar methods can be applied to 8-bit palette based images (like GIF images) but the changes, as the reader might imagine, are more dramatic[6]. This is alleviated in this paper by sorting the palette and reassigns each pixel to the index of its color in the new palette before the embedding process.

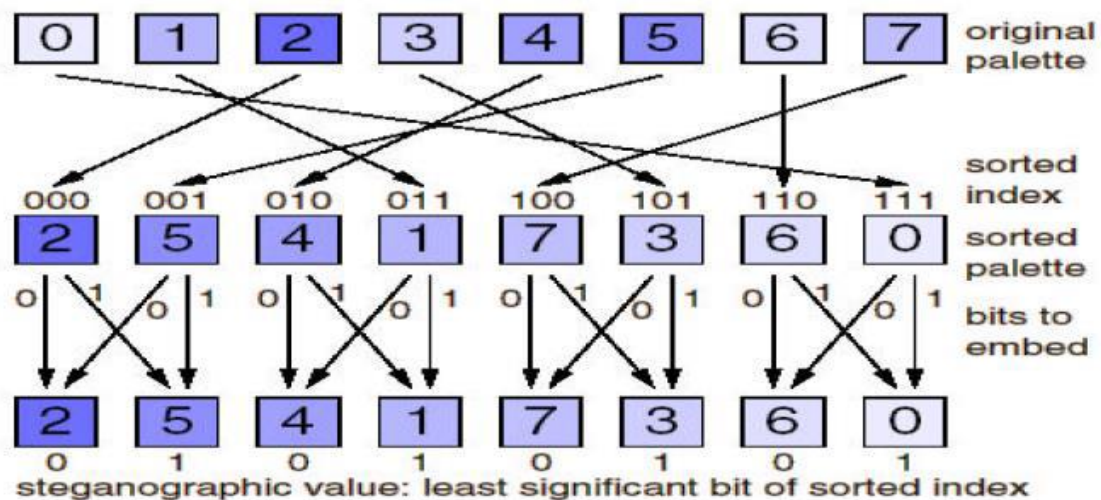


Fig 9.3: LSB Embedding

9.2. LEMPEL-ZIV-WELCH (LZW)[10]:-

LZW is a universal lossless data compression algorithm created by Abraham Lempel, Jacob Ziv, and Terry Welch. It was published by Welch in 1984 as an improved implementation of the LZ78 algorithm published by Lempel and Ziv in 1978. The algorithm is designed to be fast to implement but is not usually optimal because it performs only limited analysis of the data.

The compressor algorithm builds a string translation table from the text being compressed. The string translation table maps fixed-length codes (usually 12-bit) to strings. The string table is initialized with all single-character strings (256 entries in the case of 8-bit characters). As the compressor character-serially examines the text, it stores every unique two-character string into the table as a code/character concatenation, with the code mapping to the corresponding first character. As each two-character string is stored, the first character is sent to the output. Whenever a previously-encountered string is read from the input, the longest such previously-encountered string is determined, and then the code for this string concatenated with the extension character (the next character in the input) is stored in the table. The code for this longest previously-encountered string is output and the extension character is used as the beginning of the next word.

The decompressor algorithm only requires the compressed text as an input, since it can build an identical string table from the compressed text as it is recreating the original text. However, an abnormal case shows up whenever the sequence *character/string/character/string/character* (with the same character for each character and string for each string) is encountered in the input and *character/string* is already stored in the string table. When the decompressor reads the code for *character/string/character* in the input, it cannot resolve it because it has not yet stored this code in its table. This special case can be dealt with because the decompressor knows that the extension character is the previously-encountered *character*.

A. Compressor Algorithm

Build a table and store all possible strings in it

STRING = get input character

WHILE there are still input characters DO

CHARACTER = get input character

IF STRING+CHARACTER is in the string table then

STRING = STRING+character

ELSE

output the code for STRING

add STRING+CHARACTER to the string table

STRING = CHARACTER

END of IF

END of WHILE

output the code for STRING

B. Decompressor Algorithm

Build a table and store all possible strings in it

Read OLD_CODE

OLD_CODE = get translation of OLD_CODE

output OLD_CODE

CHARACTER = OLD_CODE

WHILE there are still input characters DO

Read NEW_CODE

IF NEW_CODE is not in the string table THEN

STRING = OLD_CODE

STRING = STRING+CHARACTER

ELSE

STRING = get translation of NEW_CODE

END of IF

output STRING

CHARACTER = first character in STRING

add OLD_CODE + CHARACTER to the string table

OLD_CODE = get translation of NEW_CODE

END of WHILE

9.3. THE WORKING SYSTEM ALGORITHMS:-

9.3.1 Embedding Process:

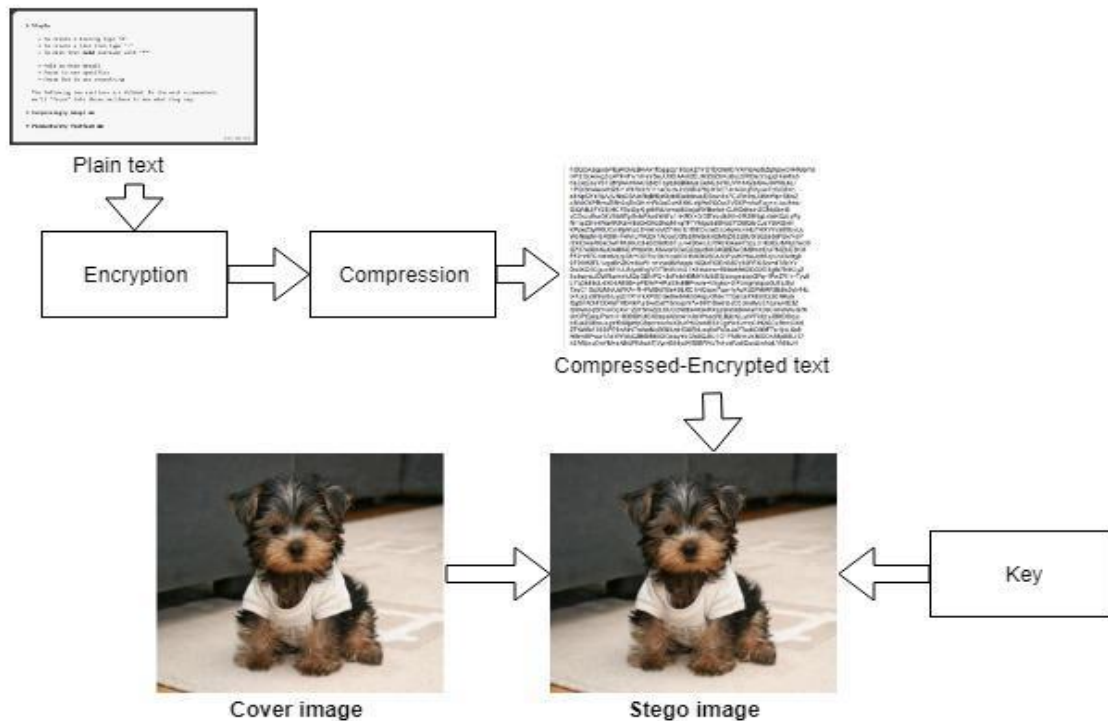


Fig 9.4: Embedding Process Block Diagram

Input: The text message and cover image represented by animated GIF/JPEG image.

Output:: The stego image represented by animated GIF/JPEG image.

Step 1: Read the text message.

Step 2: Encrypt the text message by using AES algorithm.

Step 3: Compress the encrypted message by using LZW compression algorithm.

Step 4: Extract all frames from the stegoimage (animated GIF/JPEG image) and convert them to a 256 color BMP images.

BMP images.

Step 5: For $i=1$ to no. of the bit in the LZW text code.

Step 6: For $j=1$ to no. of pixels in each frame.

Step 7: For $k=1$ to no. of image frames.

Step 8: Hide the LZW code bit in current pixel by using LSB algorithm.

Step 9: next k .

Step 10: next j .

Step 11: next i

9.3.2 Extracting Process:

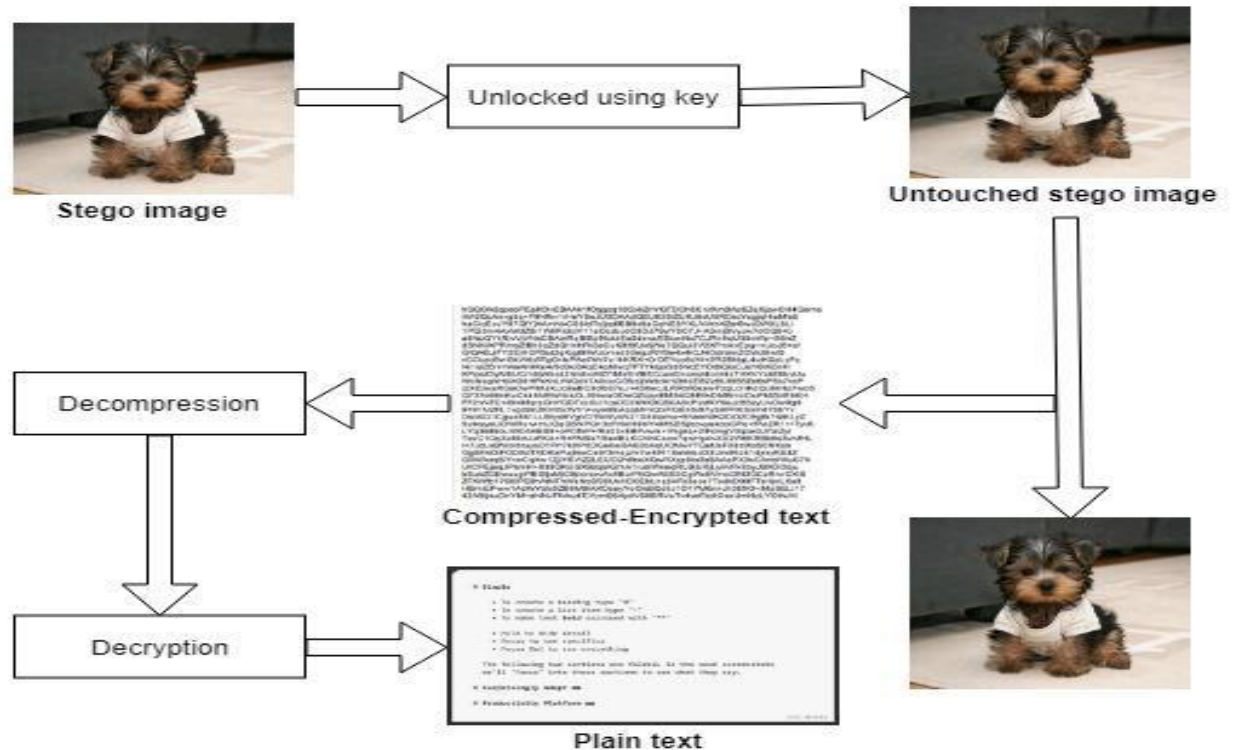


Fig 9.5: Extracting Process Block Diagram

Input: The stego image represented by animated GIF/JPEG image.

Output: The text message.

Step 1: Extract all frames from the stegoimage(animated GIF/JPEG image) and convert them to a 256 color

BMP images.

Step 2: Sort the palette of all BMP images and reassign each

Step 3: For $i=1$ to no. of the bit in the LZW text code.

Step 4: For $j=1$ to no. of pixels in each frame.

Step 5: For $k=1$ to no. of image frames.

Step 6: Extract the LZW code bit from the current pixel by Using LSB algorithm.

Step 7: next k.

Step 8: next j.

Step 9: next i.

Step 10: Decrypt the decompressed text message by using AES algorithm.

Step 11: Decompress the LZW text code by using LZW Decompression algorithm.

Chapter 10

Design Detail

10.1 Context Level Diagram

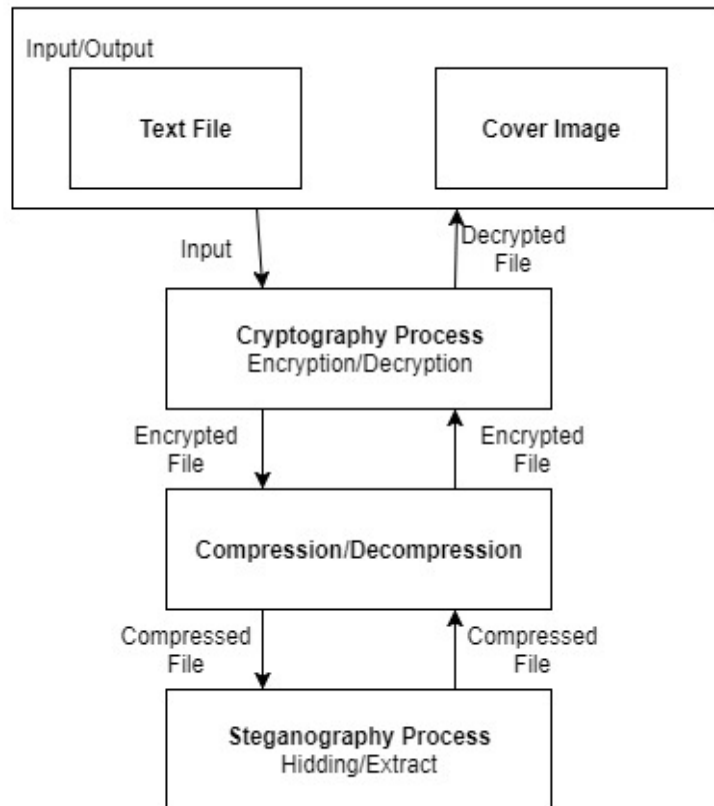


Fig 10.1: Context Level Diagram

10.2 Data Flow Diagram

Level 0:-

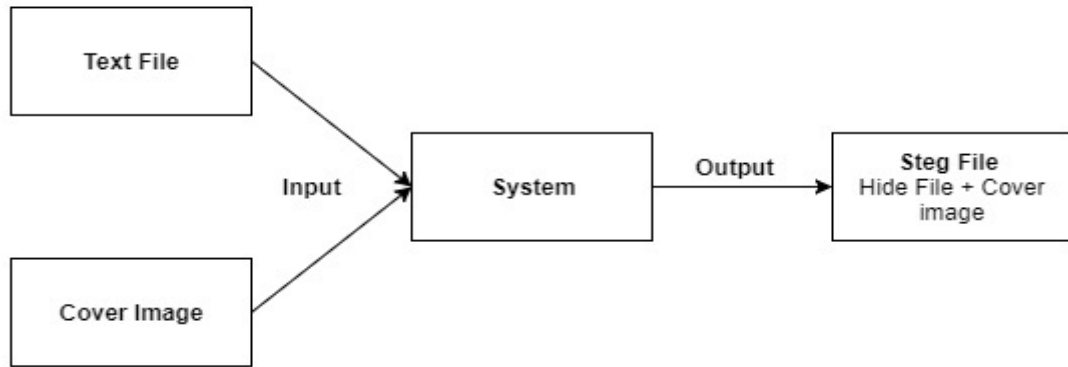


Fig 10.2.1: DFD Level 0

Level 1:-

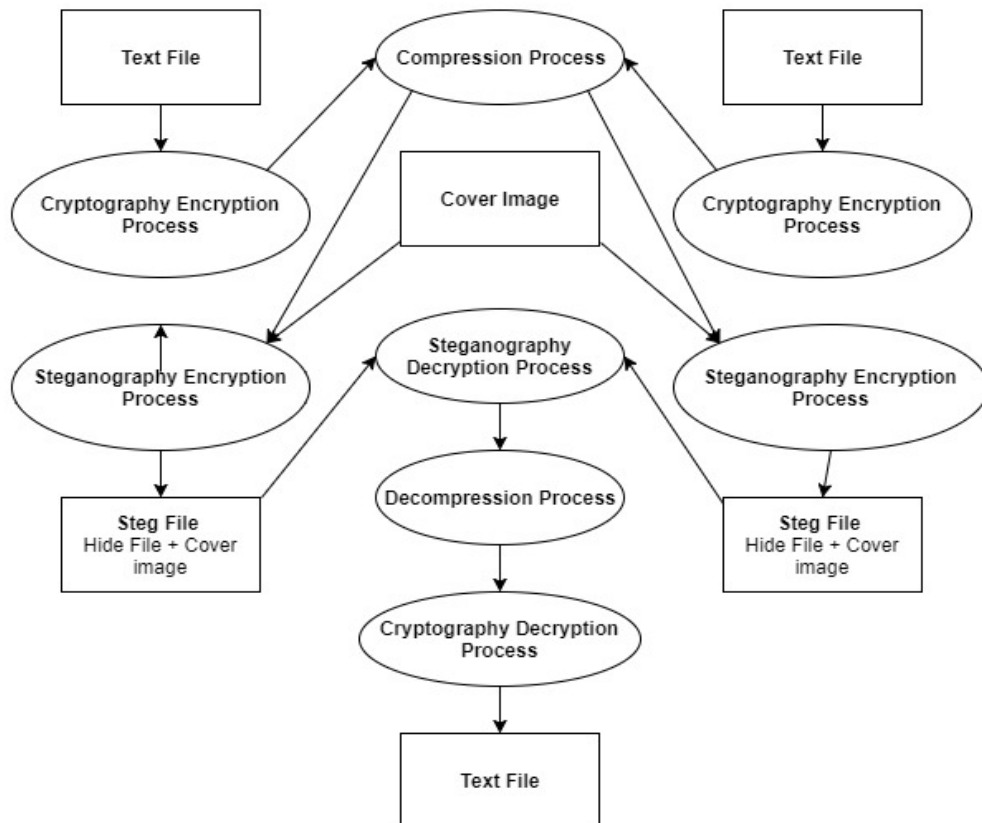


Fig 10.2.2: DFD Level 1

Level 2:-

(Cryptography process with Compression & Decompression on Text File)

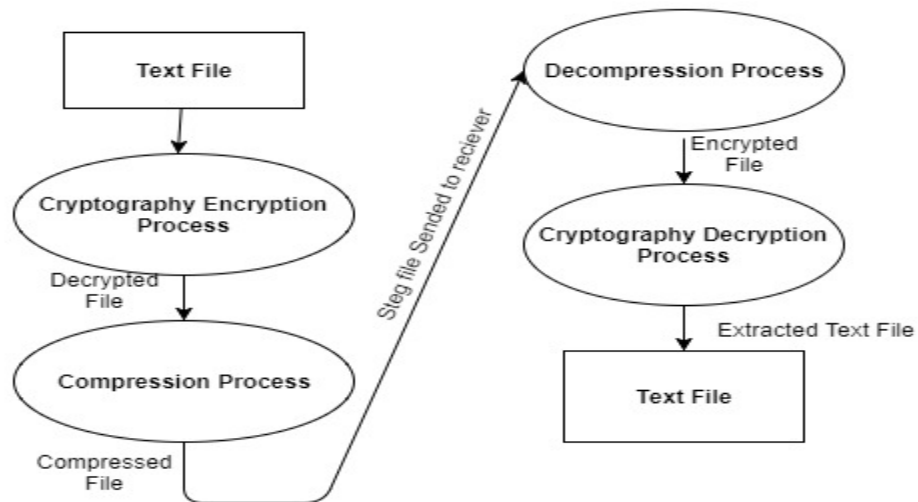


Fig 10.2.3: DFD Level 2

10.3 E-R Diagram

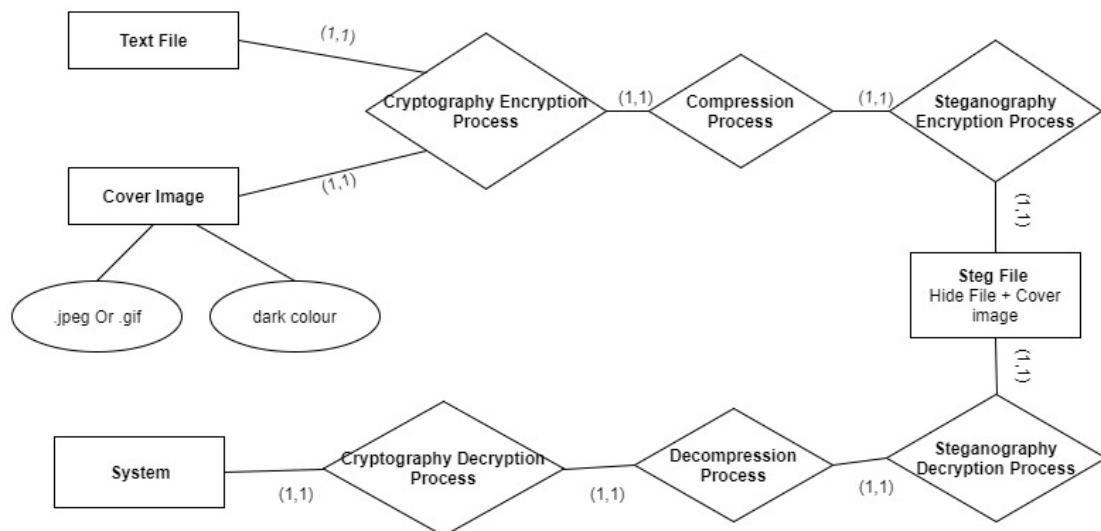


Fig 10.3: ER Diagram

10.4 Sequence Diagram

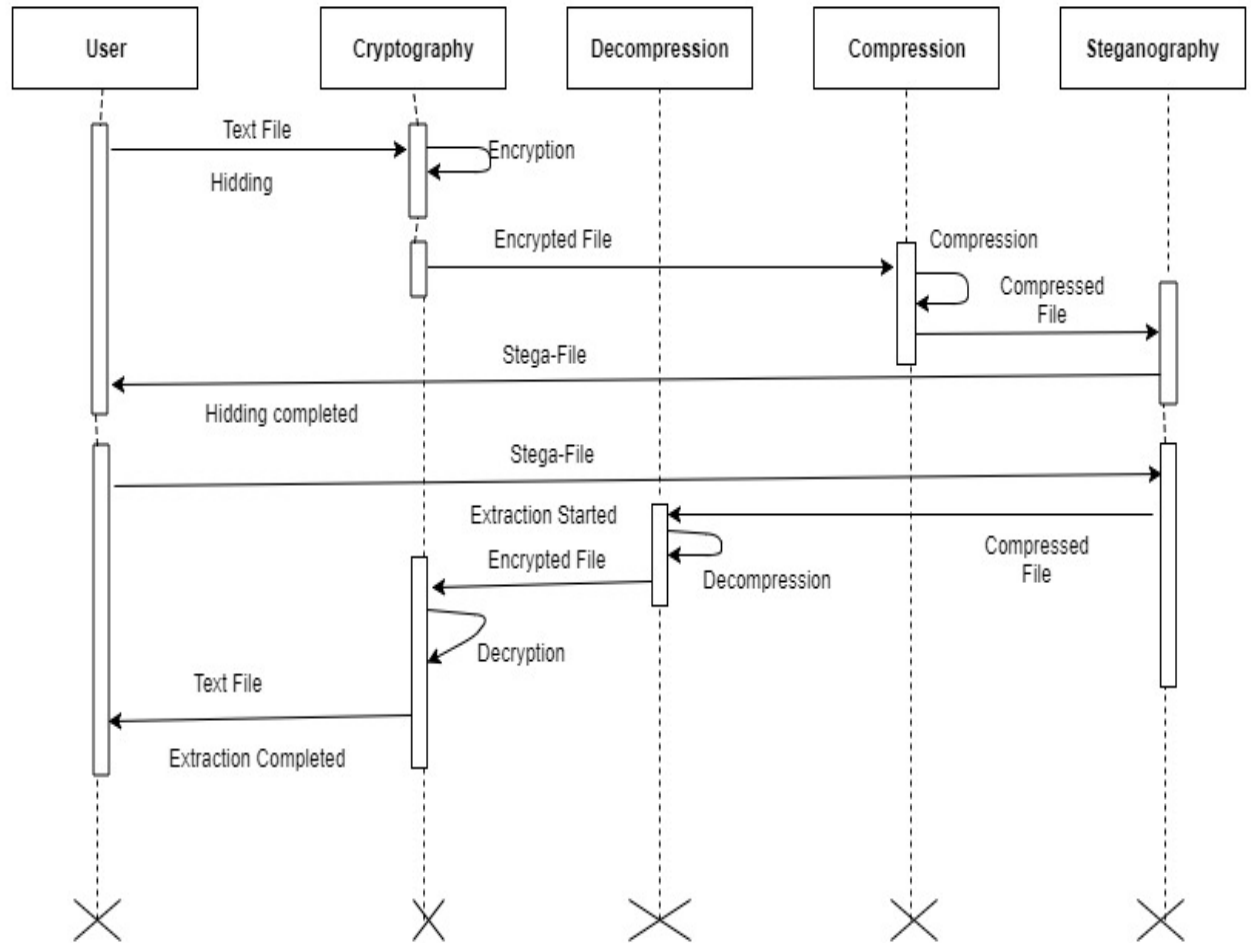


Fig 10.4: Sequence Diagram

9.5 Control Flow Diagram

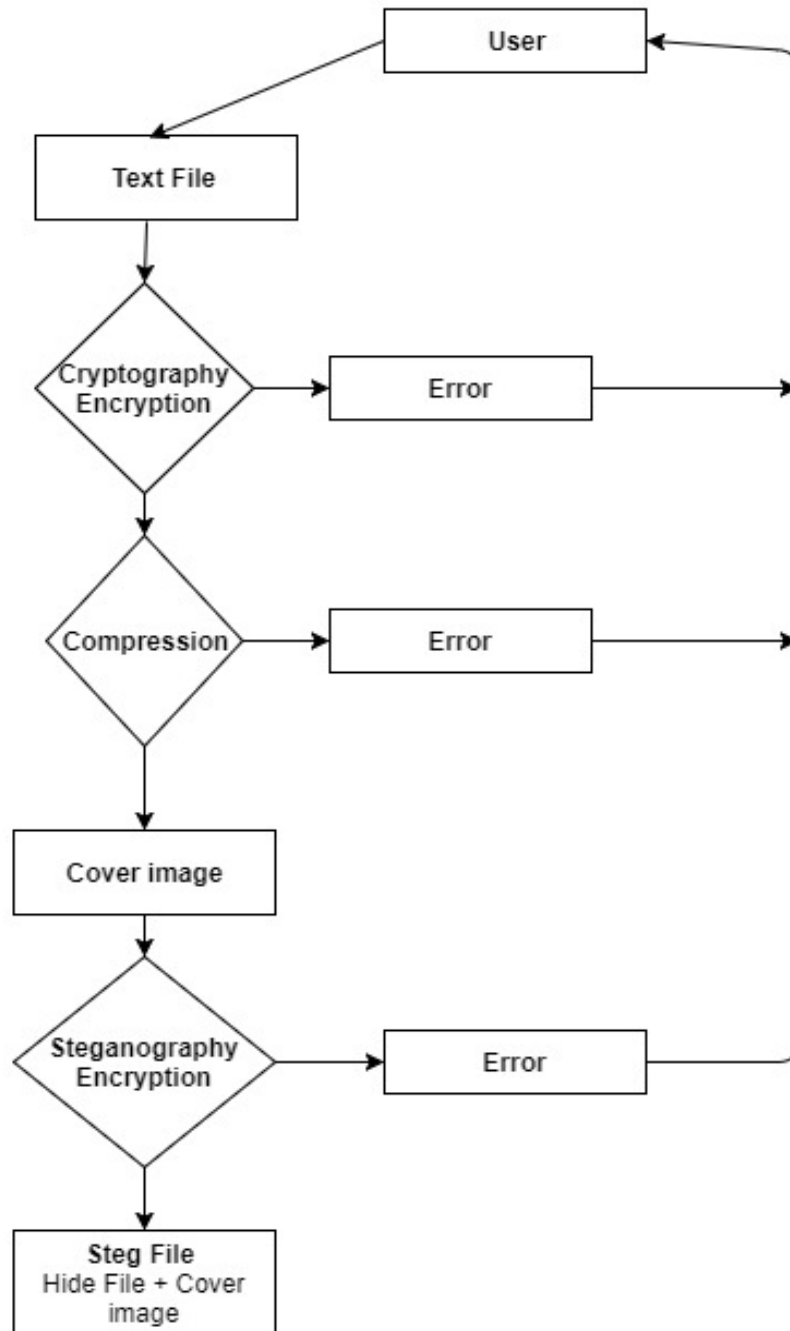


Fig 10.5: Control Flow Diagram

Chapter 11

Implementation and Experimental Set up

11.1 Hardware and Software Requirements:-

Hardware requirements: The hardware requirements for the implementation of the project are as follows:

- A system with either INTEL based or AMD based processors.
- A system with minimum of 512MB RAM
- An interfacing device to make the communication between the sensor device and the client side system possible.
- A system that will act as a server should have at least 1GB free space on hard disk.
- System should have Internet connectivity

Software requirements: The software requirements for the project are as follows:

- Windows Operating System
- Web browser
-

11.2 Data Set and Connectivity

This is the data set for login application, the local database server and the design layout of the elements of the local web host. We can see in the Figure 11.1.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	name	varchar(20)	latin1_swedish_ci		No	None			Change Drop Primary Unique Index Spatial Fulltext Distinct values More
2	email	varchar(20)	latin1_swedish_ci		No	None			Change Drop Primary Unique Index Spatial Fulltext Distinct values More
3	password	text	latin1_swedish_ci		No	None			Change Drop Primary Unique Index Spatial Fulltext Distinct values More

☐ Check all With selected:
 [Browse](#)
[Change](#)
[Drop](#)
[Primary](#)
[Unique](#)
[Index](#)
[Add to central columns](#)
[Remove from central columns](#)

Fig 11.1: Login Database

11.3 Simulation and working environment

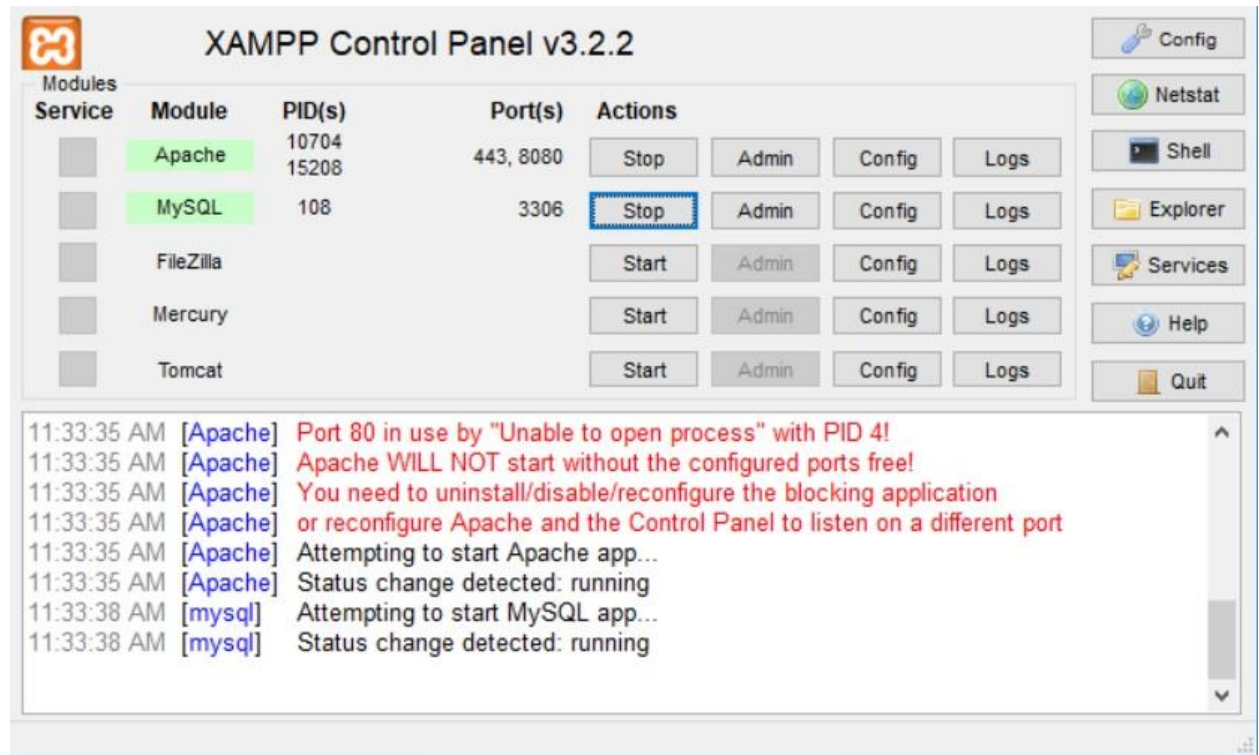


Fig 11.2: Server

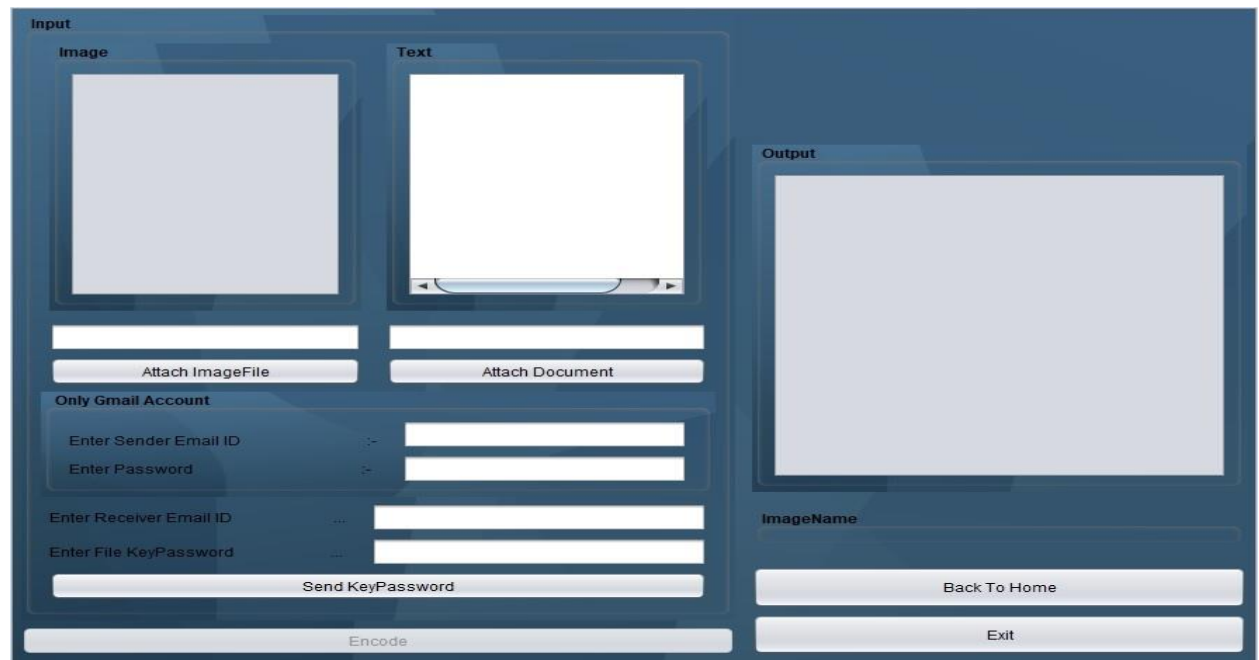


Fig 11.3: Embedding Interface

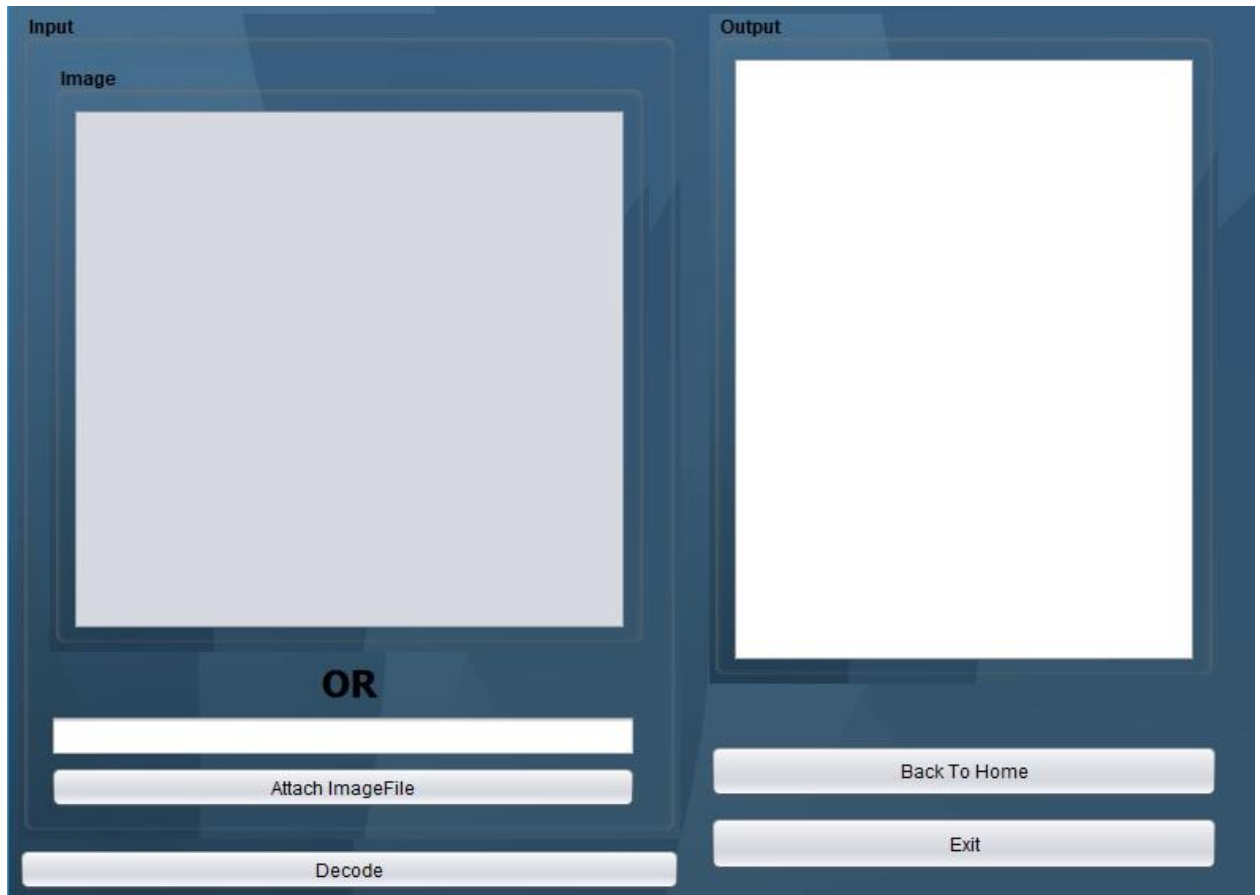


Fig 11.4: Extracting Interface

11.4 Gantt Chart

The Gantt Chart Shows planned and actual progress for a number of tasks displayed against a horizontal time scale. It is effective and easy-to-read method of indicating the actual current status for each of set of tasks compared to planned progress for each activity of the set.

Gantt Charts provide a clear picture of the current state of the project.

Table 11.1 Gantt chart

Months →	AUGUST				SEPTEMBER				OCTOBER				NOVEMBER			
Particulars ↓																
	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4

1)Gathering initial requirements																	
Decide the project topic																	
Identify needs and constraints																	
Determine goals and scope																	
Establish specification																	
Milestone : product specification complete																	
2) Feasibility																	
Technical feasibility																	
Economic feasibility																	
Application (code) feasibility																	
Operational feasibility																	
Milestone : feasibility study complete																	
3)Requirement Determination																	
Determine inputs																	
Determine outputs																	

Process control															
Synopsis															
Milestone : synopsis is ready															

Months →	DECEMBER				JANUARY				FEBRUARY				MARCH	
Particulars ↓														
	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2
4)Analysis and Design														
Phase 1														
Phase 2														
Phase 3														
Milestone : Design specification complete														
5) Development														
Developing Interface														
Developing Modules														
Milestone : Development phases complete														
6) Testing														
Testing Connectivity														
System testing and load testing														
Milestone : Testing phase complete														
7)Implementation														
Building Prototype														

Building Interface														
8) Evaluation														
Documentation and recommendation														
Milestone : Implementation and Evaluation phases complete														

Chapter 12

Testing

System testing is a critical phase implementation. Testing of the system involves hardware devise and debugging of the computer programs and testing information processing procedures. Testing can be done with text data, which attempts to stimulate all possible conditions that may arise during processing. If structured programming Methodologies have been adopted during coding the testing proceeds from higher level to lower level of program module until the entire program is tested as unit. The testing methods adopted during the testing of the system were unit testing and integrated testing

12.1 Unit Testing

Unit testing focuses first on the modules, independently of one another, to locate errors. This enables the tester to detect errors in coding and logical errors that is contained within that module alone. Those resulting from the interaction between modules are initially avoided. Tested on following modules:

- Login - On entering correct details a user is able to login successfully
- Embedding and extraction process is working successfully.

12.2 Integration Testing

Integration testing is a systematic technique for constructing the program structure while at the same time to uncover the errors associated with interfacing. The objective is to take unit-tested module and build a program structure that has been detected by designing. It also tests to find the discrepancies between the system and its original objectives. Subordinate stubs are replaced one at time actual module. Tests were conducted at each module was integrated. On completion of each set another stub was replaced with the real module. Interfaces tested on:

- Login interface :- User logs in and perform embedding and extraction process successfully.
- Steganography interface :- Hiding and extraction process work successfully.

- Cryptography interface :- Encryption and Decryption by using secret encryption key is working successfully.
- Compression/decompression interface :-Compression and Decompression by using LZW compression algorithm is working successfully.
- Multi-threading approach for fast extraction from stego-image is working successfully.
- Secret key mailing through Gmail to receiver is also working successfully.

12.3 Functional Testing

Functional testing is a technique in which all the functionalities of the program are tested to check whether all the functions that were proposed during the planning phase is full filled. This is also to check that if all the functions proposed are working properly. This is further done in two phases:

- One before the integration to see if all the unit components work properly
- Second to see if they still work properly after they have been integrated to check if some functional compatibility issues arise. Functionalities for user, admin, file access embedding and extraction was tested and executed to utmost satisfaction.

12.6 Test Cases

1. Figure 12.1, 12.2 and 12.3 displays the test case conducted to authenticate and validate the user for granting server access by entering correct login credentials.

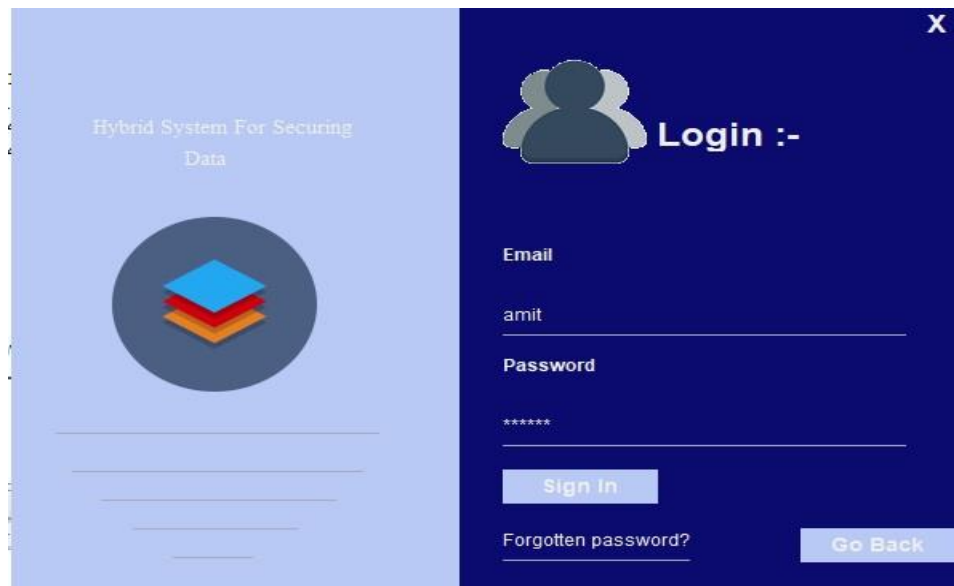
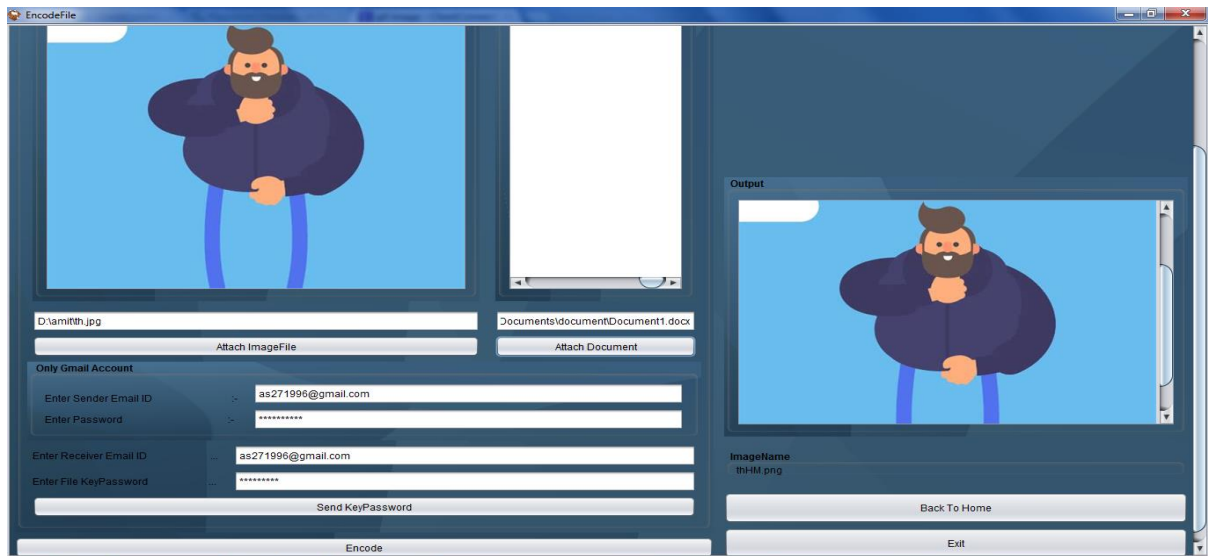
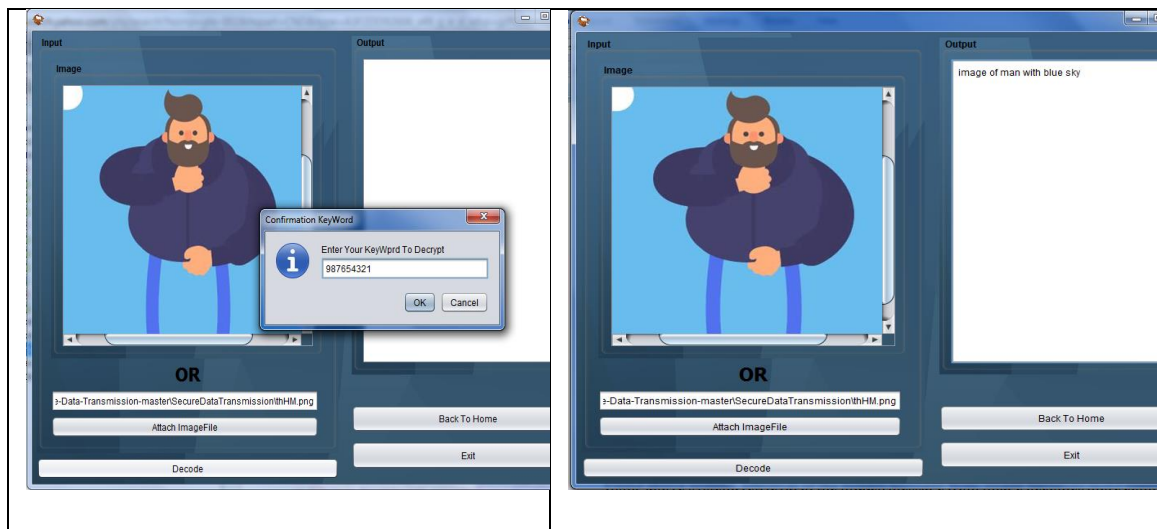


Fig 12.1: Login Test

**Fig 12.2: Embedding Test****Fig 12.3: Extracting Test**

2. Figure 12.4,12.5 displays the test case conducted to verify that invalid user credentials does not get access.

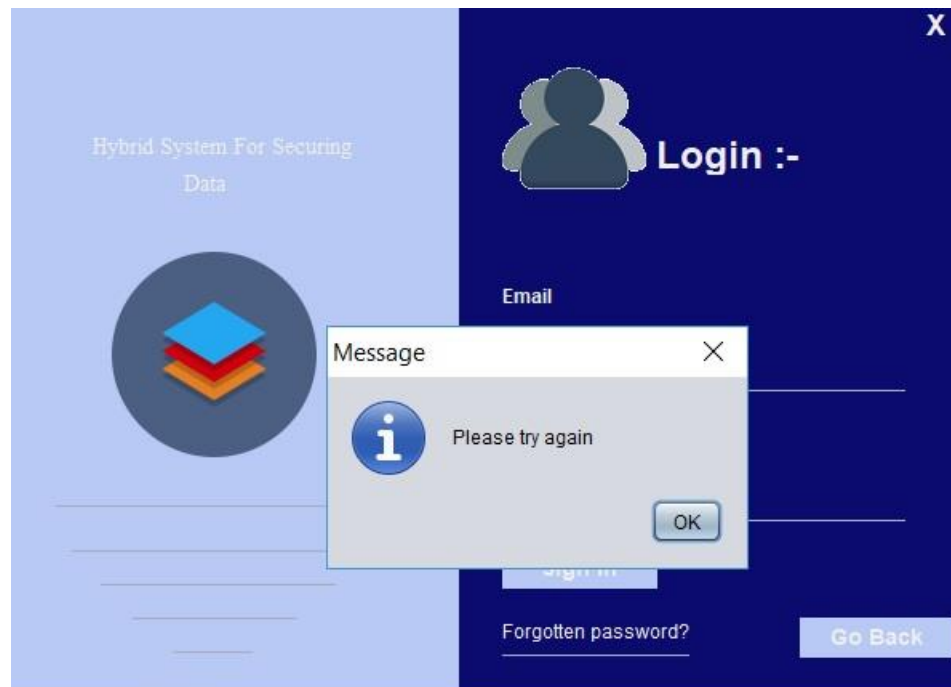


Fig 12.4: Test Case for invalid login credentials



Fig 12.5: Test Case for invalid credentials for extraction

PSNR

The Peak Signal to Noise Ratio (PSNR) is the ratio between maximum possible power and corrupting noise that affect representation of image. PSNR is usually expressed as decibel scale. The PSNR is commonly used as measure of quality reconstruction of image. The signal in this case is original data and the noise is the error introduced. High value of PSNR indicates the high quality of image.

$$\text{PSNR1} = 10 * \log_{10}((255)^2 / \text{MSE})$$

Here MSE is a mean square error

MSE

Mean Square Error can be estimated in one of many ways to quantify the difference between values implied by an estimate and the true quality being certificated. MSE is a risk function corresponding to the expected value of squared error. The MSE is the second moment of error and thus incorporates both the variance of the estimate and its bias. The MSE of an estimate and is defined as

$$\text{MSE} = 2$$

Capacity

The maximum amount of information that can be carried in the media, steganography algorithm can be implanted without being carried in the media to apply tangible change. High capacity steganography algorithms is to evaluate the main parameters. Due to compression algorithm hiding capacity is increases by double.

Chapter 13

Result and Analysis

This section outlines the work flow of the system and analyzing the end result of the system architecture.

Logging in the System: Initially a user must login the system using the correct credentials. Upon authorization by the system; the user is directed to the default home page termed DASHBOARD

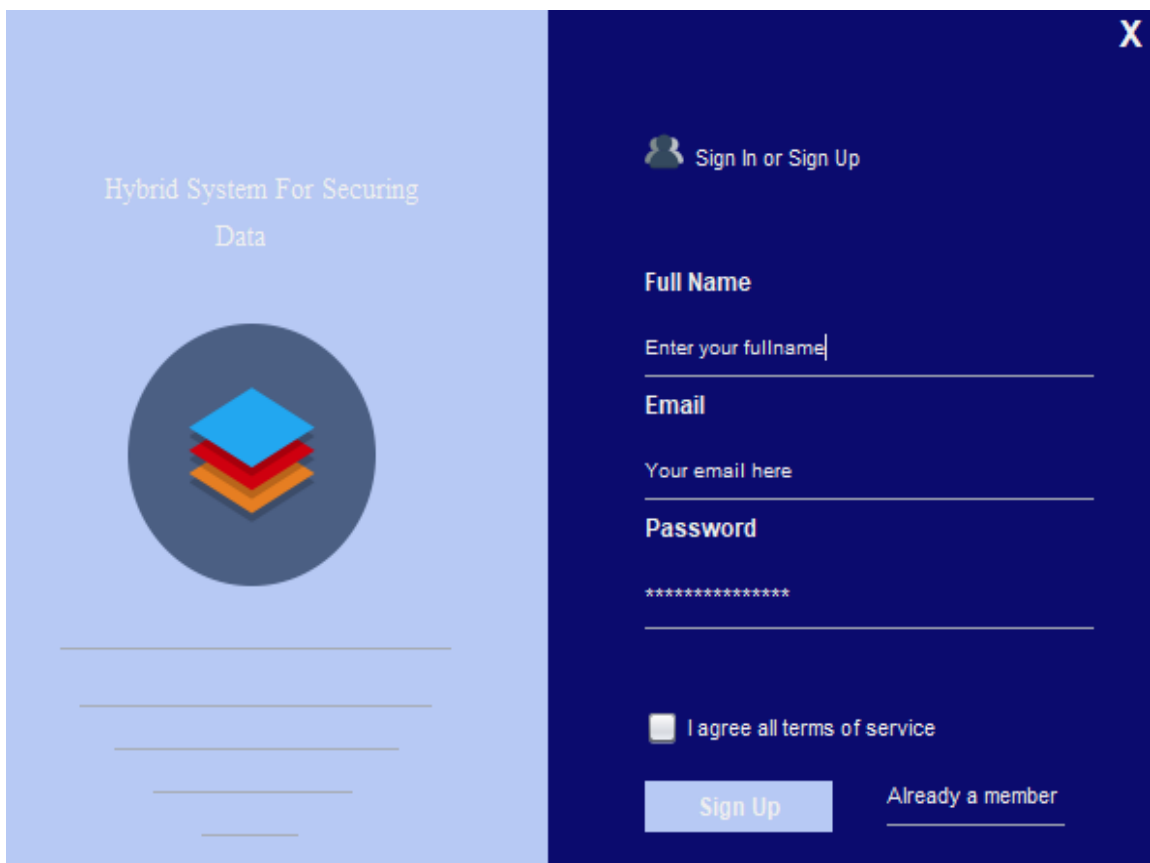
The image shows a web registration interface. The left side has a light blue background with the text 'Hybrid System For Securing Data' and a logo consisting of three stacked squares (blue, red, orange) inside a dark blue circle. Below the logo are four horizontal lines for input. The right side has a dark blue background with a close button 'X' in the top right corner. It features a 'Sign In or Sign Up' link with a user icon, followed by input fields for 'Full Name' (placeholder: 'Enter your fullname'), 'Email' (placeholder: 'Your email here'), and 'Password' (placeholder: '*****'). Below the password field is a checkbox labeled 'I agree all terms of service'. At the bottom are two buttons: 'Sign Up' and 'Already a member'.

Fig 13.1: Registration Interface

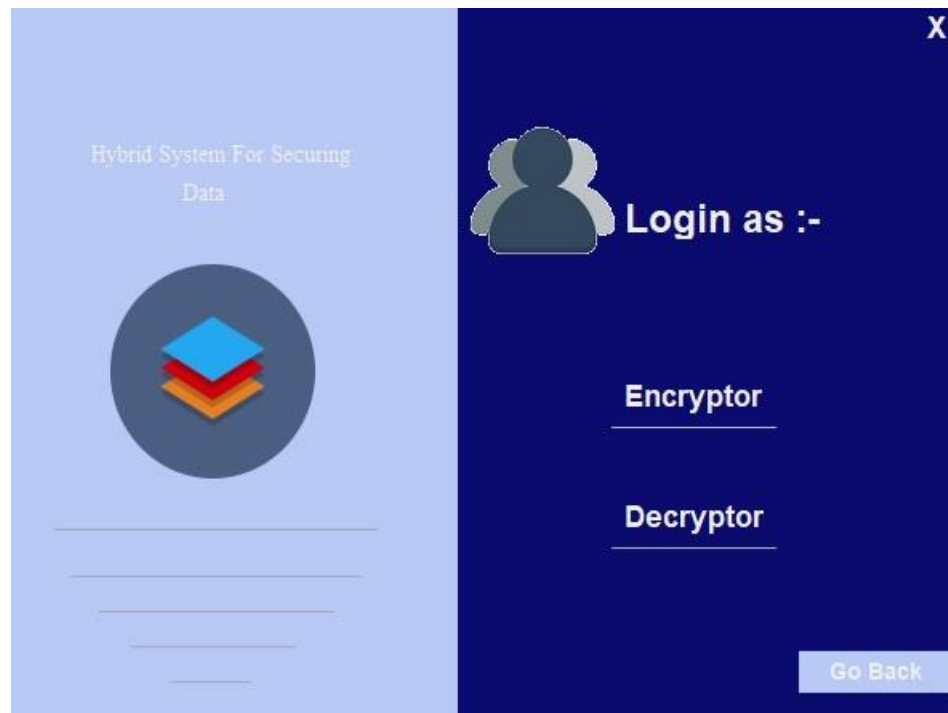


Fig 13.2: Login choice Interface

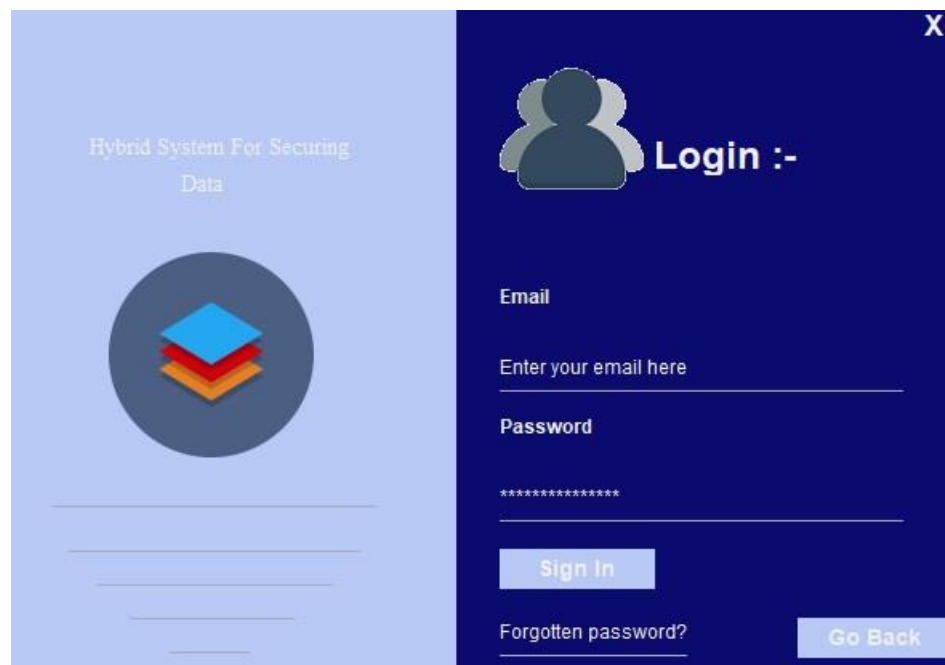


Fig 13.3: Login Interface

Embedding Process

The screenshot shows a web application interface with a dark blue background. It is divided into three main sections: Input, Text, and Output. The Input section contains a large white square placeholder for an image and a smaller white square placeholder for text. Below these are two buttons: 'Attach ImageFile' and 'Attach Document'. The Text section contains a large white square placeholder for text. Below this is a button labeled 'Encode'. The Output section contains a large white square placeholder for the output. Below this is a button labeled 'Back To Home'. There is also a section labeled 'Only Gmail Account' with fields for 'Enter Sender Email ID', 'Enter Password', 'Enter Receiver Email ID', and 'Enter File KeyPassword', followed by a 'Send KeyPassword' button. At the bottom right, there is a button labeled 'Exit'.

Fig 13.4: Successful login as encryptor

Two images are shown in Fig.7 which explain the embedding process of data. Here we obtained stego-image which exactly looks like cover image. And also stego-image is a lock with the key provided by the user which is directly transferred to the receiver through mail- in an encrypted format. To use that key, user needs to first decrypt that key and then use for the extraction process.

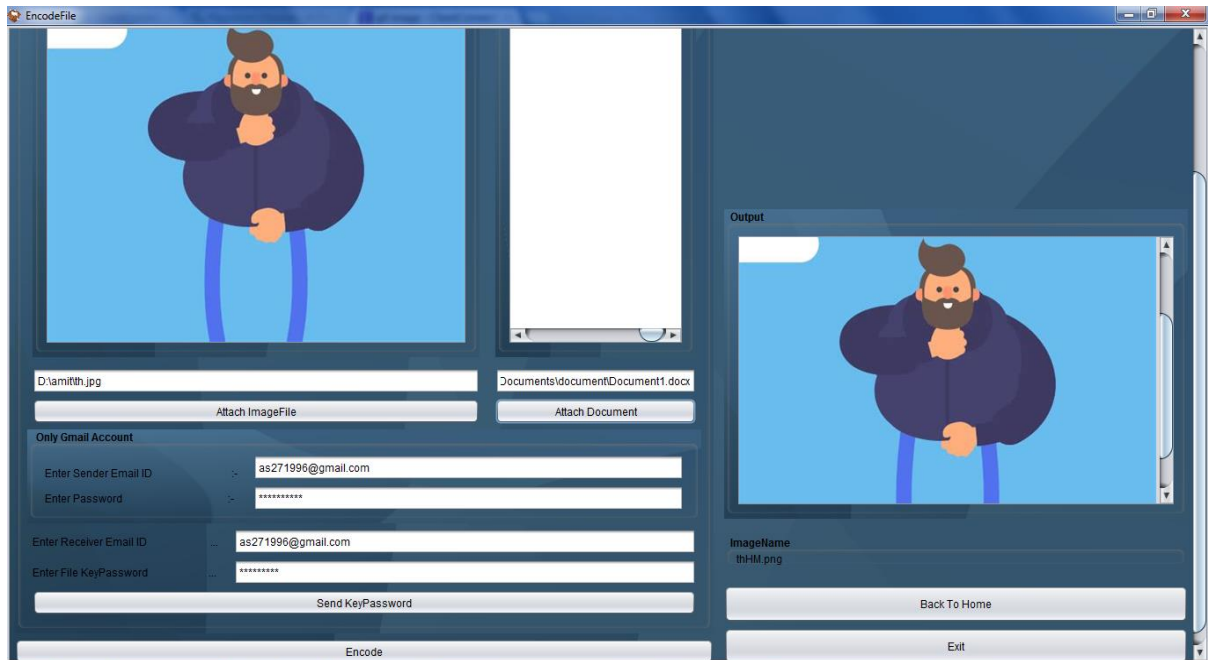


Fig 13.5: Embedding data

Compression/decompression

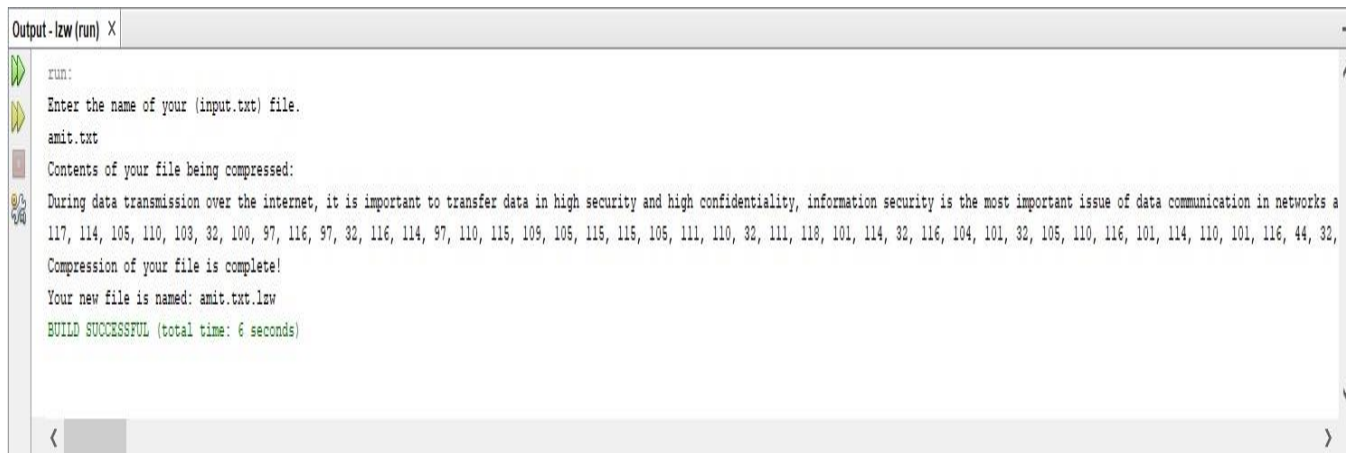


Fig 13.6: Index of LZW compression

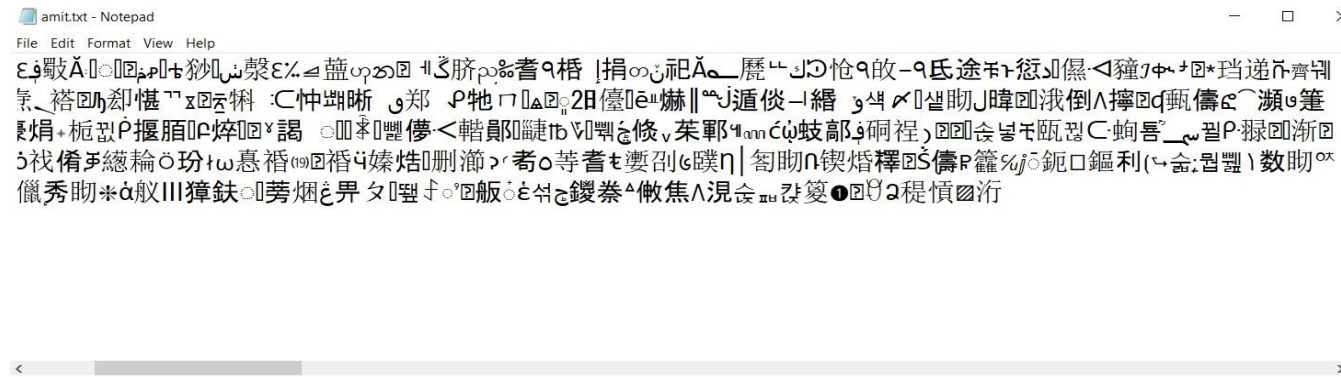


Fig 13.7: Compressed file

During data transmission over the internet, it is important to transfer data in high security and high confidentiality, information security is the most important issue of data communication in networks and internet. To secure transferred information from intruders, it is important to convert information into the cryptic format. Different methods are used to ensure data security and confidentiality during transmissions like Steganography and cryptography. Proposed work improves information security through developing efficient compression of texts and encryption of texts by cryptography with steganography. The Proposed algorithm ensures the encryption and decryption using AES, compression and decompression by using LZW and RGB pixel shuffling with steganography. LSB method is used to insert data bits in LSB of RGB pixels of the cover image. In this Multi-threading use to retrieve information from image rapidly. Hence proposed system can reduce the data transmission time and cost. These algorithms are performed by using JAVA program.

Fig 13.8: Decompressed file

Extraction Process:

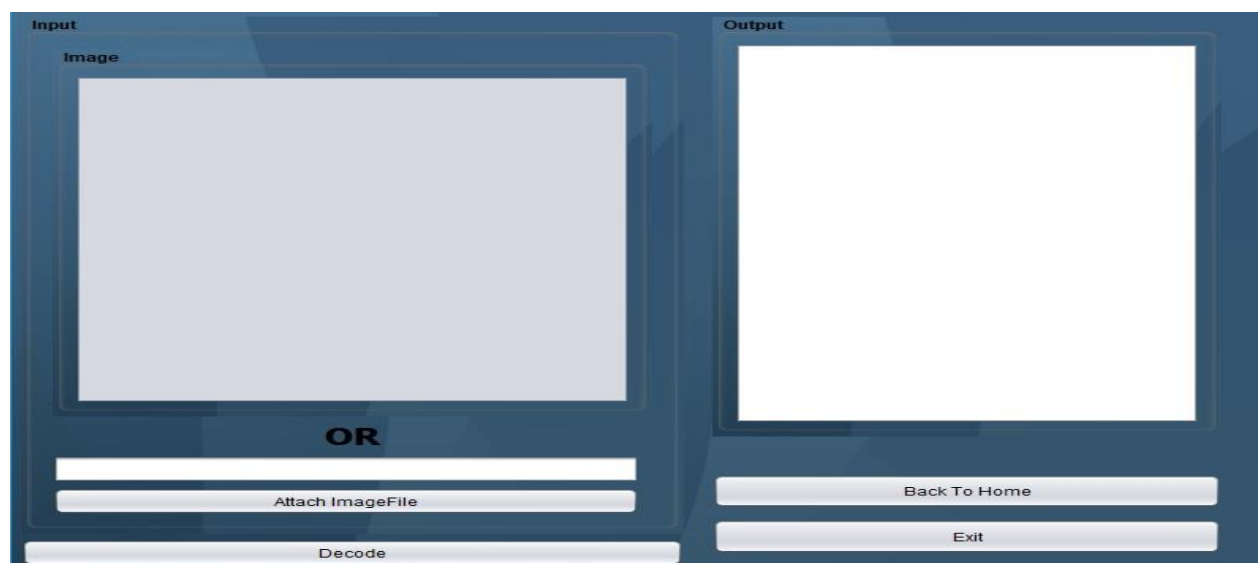


Fig 13.8: Successful login as decryptor

In Fig.8 there is two column, in that first column contain stego-image which requires the key, provided by the sender through the mail, for extraction of information from the stego image. And in another half of column shows the acceptance of key for completing the extraction process.

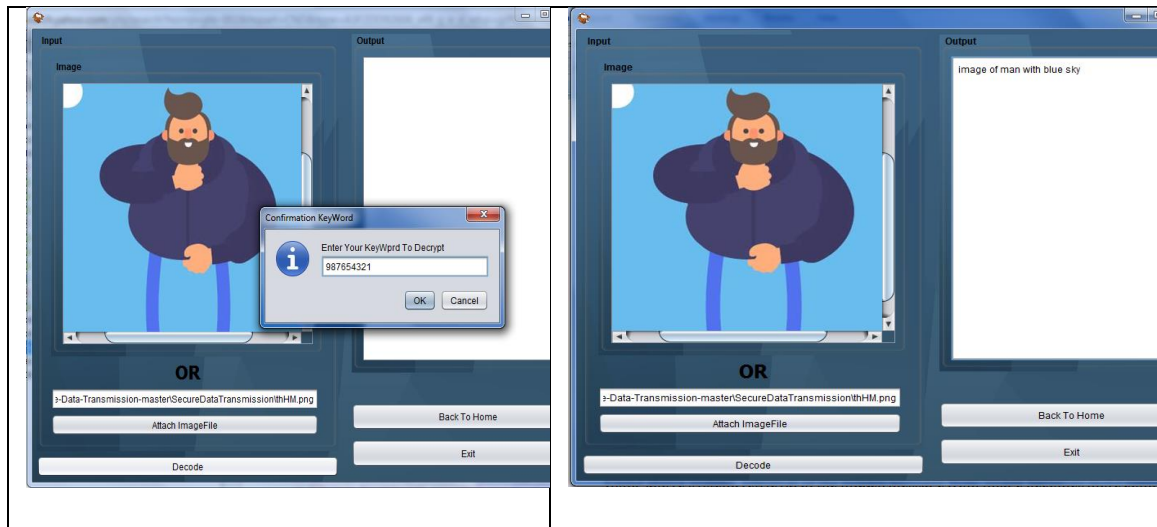


Fig 13.9: Extracting Data

Chapter 14

Advantages and Limitations

Advantages

- The ability of the system to generate Stego-image with the combination of cryptography, and compression technique and offer same/more level of security in comparison with the already existing technologies.
- Owing to generation of stego-image, the proposed system generates less overhead as compared to the earlier Stego-system.
- Proposed system utilizes less system power and consumes very less system memory without compromising on the security front.
- All advantages combined together Steganography is the most suitable form of securing data suited especially for portable and embedded devices.
- Leakage of data by authorized administrators and even by management can be easily recorded without having to compromise the integrity or confidentiality of the data.
- It has various layers of security so that stealing data is very difficult.
- If intruder steals data from the network then also extraction of data from stego-image will be very much difficult because of key which is locked the stego-image, compression which has its own index file and also from cryptography which has a secret key for encryption and decryption.

Limitation

- For steganographic communication to function, the recipients must have some reason to look for hidden messages
- Steganographic communication often requires a “shared secret.” Such shared secrets may be when and where to look for hidden messages, and how to extract hidden information once obtained. This information must remain secret; otherwise the goal of hidden communication is defeated

- Once an image has been used to hide a message, the original should be destroyed. Otherwise, someone may notice the difference between the original and stego-image.
- The biggest limitation observed in the system and as suggested by researchers and scientists is the minimalistic exploration of the proposed domain.

Chapter 15

Future Scope

In our randomized LSB technique we have embedded two bits in a pixel using a message dependent randomized approach. In future we would like to exploit the possibility of hiding three bits in moderate bit locations in a randomized manner without disturbing the least significant bit in each pixel. The possibility of embedding by changing every pixel value with a new one which will conceal the hidden data can also be exploited. LSB substitution methods give high capacity, whereas PVD methods give high security. Some techniques are already available which combines both the LSB and PVD approaches to get both high capacity and high security. However, better techniques are always warranted through the combination of both LSB and PVD approaches which can address the short-comings in the previously existing techniques. By using better encryption algorithm we can improve the security level and also using better lossless compression algorithm, we can compress file as much we can, by doing this we will reduce cost of transmission and also reduce transmission time over the network and increases hiding capability of stego-image.

Chapter 16

Conclusion

Using above system, retrieval of the hidden message from image becomes more complex because text messages are compressed using LZW which makes processing difficulties for the intruder. another advantage of the system is that, for any observer, it becomes difficult to recognize difference between original image and image with hidden text message compared to other approaches. Here the lossless compression method used in the image and in the embedded text result to extract the text without any changes in the message. Other cryptographic algorithms, compressions methods can be used with the proposed system in future to improve performance.

Chapter 17

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